

Estimating the Gravity Equation

PPML, internal trade, and staggered trade agreements

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Where we stopped

Theory told us what to control for. In a panel:

$$\ln X_{ijt} = f_{it} + f_{jt} + \kappa \tau_{ij} + \epsilon_{ijt}$$

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Three problems remain, and each has been the subject of a literature.

- 1 Estimating in **logs** deletes the **zero flows**, and under **heteroskedasticity** this biases the estimate.
- 2 We do not take **internal trade flows** into account.
- 3 We treat an **agreement** as if it were a clean experiment.

Section 1

Why PPML

Problem 1 — the zeros

Taking logs requires $X_{ijt} > 0$, and $\ln 0$ is undefined. Faced with this, the usual practice is either to **drop** the zero observations, or to estimate $\ln(1 + X_{ijt})$ — adding one dollar of trade to every pair in the world.

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Zeros are abundant. A large share of country pairs record no trade at all in a given year. They are not a handful of awkward observations.

Zeros are informative. A zero is the outcome we most want to explain: it tells us that trade costs were high enough that *no* firm found it profitable to serve that market. Understanding why two countries do not trade is part of understanding trade.

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Helpman et al. (2008) build a model in which the decision to trade at all — the **extensive margin** — is separate from how much is traded, and show that ignoring the zeros biases the estimated effect of trade barriers.

The zeros are not missing at random

Small, distant and poor pairs are the ones that do not trade. They are also exactly the pairs on which the effect of distance is most visible.

Selection on the dependent variable

Dropping them selects the sample on the outcome itself, which makes the estimator inconsistent. Adding one before taking the log is worse: the answer then depends on the unit in which trade happens to be measured.

Problem 2 — logs and heteroskedasticity

The theory gives a **multiplicative** model, $X_{ij} = \exp[\cdot] \epsilon_{ij}$. We estimate its log. But

$$\mathbb{E}[\ln \epsilon] \neq \ln \mathbb{E}[\epsilon]$$

- Taking the log of the expectation is not the expectation of the log (Jensen's inequality).
- Trade data are strongly **heteroskedastic**: the variance of ϵ_{ij} depends on the regressors — large economies and close neighbours have both bigger flows and more dispersed ones.
- Under **heteroskedasticity**, $\mathbb{E}[\ln \epsilon_{ij} \mid X]$ therefore depends on X as well, and the log-linear OLS estimator is **inconsistent** — not merely inefficient.

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This is the central argument of Santos Silva and Tenreyro (2006): the coefficients of a log-linearised model estimated by OLS cannot be read as elasticities in the presence of heteroskedasticity. They find OLS roughly **doubles** the estimated effect of distance and colonial ties. See also Head and Mayer (2014) and Fally (2015).

The answer: estimate in levels

PPML

$$X_{ijt} = \exp [f_{it} + f_{jt} + \kappa \tau_{ij} + \alpha] \epsilon_{ijt}$$

Poisson pseudo-maximum likelihood, estimated on trade **in levels**.

- **Zeros are kept.** $X_{ijt} = 0$ is a perfectly good observation.
- **Consistent under heteroskedasticity**, whatever the true distribution of ϵ . Nothing here assumes trade is Poisson-distributed — only that the conditional mean is right.
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Coefficients are now read as semi-elasticities: the effect on trade of a dummy $\hat{\kappa}$ is $\exp(\hat{\kappa}) - 1$.

Section 2

How this changes the results

The RTA results

The RTA dummy, estimated five ways on the same CEPII data.

	(1) Naive	(2) Naive EU	(3) TWFE	(4) Bil. FE	(5) PPML
RTA dum.	0.848 ^a (0.034)	0.724 ^a (0.036)	0.488 ^a (0.034)	0.290 ^a (0.026)	−0.030 (0.049)
Year FE	✓	✓	—	—	—
Exporter-year FE	—	—	✓	✓	✓
Importer-year FE	—	—	✓	✓	✓
Country-pair FE	—	—	—	✓	✓
Estimator	OLS	OLS	OLS	OLS	PPML

Dependent variable in logs in (1)–(4), in levels in (5). ^a $p < 0.01$.

You will find on Elearn an exercise and all the material to reproduce that table in R.

Reading that line

Each step removes something the previous column was wrongly attributing to the agreement.

- $0.848 \rightarrow 0.488$: the **multilateral resistances**. Part of what looked like an RTA effect was the members' general remoteness.
- $0.488 \rightarrow 0.290$: the **pair fixed effects**. Part of the rest was the time-invariant reason those two countries were close partners to begin with — Baier and Bergstrand (2007).
- $0.290 \rightarrow -0.030$: **PPML**. The log-linear estimate was driven by the small flows that dominate a log regression.

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An uncomfortable result

Read literally, column (5) says trade agreements do nothing. Hold that thought — we will see that this specification is itself biased, and biased *downwards*.

Section 3

Internal trade flows

The diagonal of the trade matrix

X_{ii} is what a country's producers sell to its own consumers.

- For a long time gravity regressions used **international flows only** — not by choice, but because internal trade was simply **not in the databases**. Customs record what crosses a border; what a country sells to itself crosses none.
- Yet X_{ii} is usually the **largest flow in the data**: most of what a country consumes, it produces at home.

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Two reasons why throwing it away is costly. Neither is about sample size.

Reason 1 — frictions need a benchmark

A border effect has no natural zero.

- “Trade between Canadian provinces is 22 times larger than trade between provinces and US states” is a statement about **internal versus international** trade.
- Without the diagonal, there is nothing to measure the border against.

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- Without the diagonal, there is nothing to measure the border against.

This is exactly the comparison behind McCallum (1995) and its resolution by Anderson and Van Wincoop (2003). Yotov (2022) shows that adhering to the theory by including domestic flows materially changes gravity estimates.

Reason 2 — non-discriminatory policies vanish

Consider a policy that moves a country's trade with **every** partner at once: WTO membership, a uniform tariff, a wave of globalisation.

- With international flows only, such a policy is **indistinguishable from an exporter-year fixed effect**. The fixed effect absorbs it entirely.
- Adding domestic trade provides the within-country control that breaks the collinearity: the policy is identified against the country's **own internal sales**.

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Heid et al. (2021) formalise this. It is the mechanism behind the resolution of the WTO puzzle: Larch et al. (2025) recover an effect of about +171% where Rose (2004) had found nothing.

Consequences

- Dai et al. (2014): omitting internal trade biases the estimated effect of regional agreements, which divert trade away from members' domestic sales.
- Including domestic flows is now a standard recommendation (Yotov et al. 2016; Yotov 2022), and the structural gravity databases distributed for research report both kinds of flow.

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A caveat on the table we just read

The CEPII sample used for that table contains *no* internal trade. This is one reason its GATT/WTO coefficient is so small.

Section 4

Trade agreements as a staggered experiment

The agreement as a treatment

Writing $RTA_{ij,t}$ splits the sample in two: pairs with an agreement in year t are treated, every pair with a zero is a control.

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Two problems survive. They are different, and the second is the one nobody saw for thirty years.

Problem 1 — who is the control group?

The control group is not a neutral benchmark. It is dominated by pairs that never sign anything.

In the sample of Nagengast and Yotov (2025), never-treated pairs lie on average **1,400 km further apart** and share a border **three times less often** than treated pairs.

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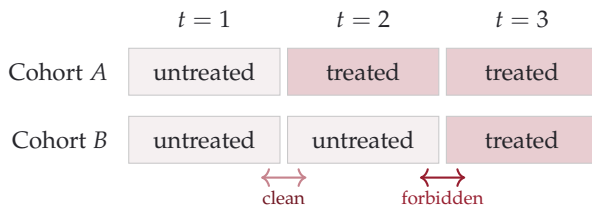
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- Pair fixed effects absorb these differences in **levels** — which is why they are indispensable.
- They say nothing about **trends**.

Identification requires that trade between two countries which never sign would have grown at the same rate as trade between future members. Countries do not sign at random, so the **parallel trends** assumption may fail, and the estimate with it.

Problem 2 — treatment starts at different dates

The EEC pairs are treated from the late 1950s, the NAFTA pairs from 1994. A pair treated late must be compared with pairs whose treatment is **not changing** at that moment — and some of those have been treated for years.



Between $t = 2$ and $t = 3$, cohort B switches on and cohort A serves as its control — but A's own effect is still growing.

The bias, made visible

Two equal cohorts, three periods, true effects $\delta_{A2}, \delta_{A3}, \delta_{B3}$. Under parallel trends and no anticipation:

$$\delta^{\text{TWFE}} = \delta_{A2} - \frac{1}{2}\delta_{A3} + \frac{1}{2}\delta_{B3} \quad \delta^{\text{ETWFE}} = \frac{1}{3}\delta_{A2} + \frac{1}{3}\delta_{A3} + \frac{1}{3}\delta_{B3}$$

- TWFE puts a weight of $-\frac{1}{2}$ on δ_{A3} — the **mature** effect of the early cohort.
- It therefore **overweights short-run effects** and **underweights long-run ones**.

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Chaisemartin and D'Haultfœuille (2020) and Goodman-Bacon (2021) show this in general: TWFE is a weighted sum of cohort-by-year effects, and some weights are negative. The estimate need not even lie inside the range of the true effects.

The fix — extended two-way fixed effects

Do not collapse the effect into one coefficient before estimating it. Saturate the model with cohort $\times$ year interactions (Wooldridge 2023):

$$Y_{ij,t} = \exp \left[\sum_g \sum_{s \geq g} \delta_{gs} D_{gs}^{ij} + f_{it} + f_{jt} + \mu_{ij} + \theta_{ii,t} \right] \epsilon_{ij,t}$$

- $D_{gs}^{ij} = 1$ if pair ij signed in year g and the year is $s \geq g$.
- Each δ_{gs} uses only **clean comparisons**, against not-yet-treated and never-treated pairs.
- Aggregate afterwards, with weights you choose and can see:
 $\hat{\delta} = \sum_{g,s} \frac{N_{gs}}{N_D} \hat{\delta}_{gs}.$

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Averaging over cohorts instead of years gives an **event study** of how the effect builds up.

What changes

Same data, same gravity best practice, **only the treatment specification changes.**

	TWFE	ETWFE
RTA coefficient	0.166*** (0.050)	0.381*** (0.041)
Implied trade effect	+18%	+46%

Nagengast and Yotov (2025), WTO Structural Gravity Database, manufacturing, 1980–2016, 69 countries, with internal trade. *** $p < 0.01$.

The gap is the downward bias that forbidden comparisons and the overweighting of short-run effects impart to the conventional estimate.

Two lessons to keep

The effect is spread over a longer horizon. Agreements keep raising trade for many years — exactly the maturation that a static TWFE compresses into the early years.

When treatment starts is a modelling choice. Because anticipation contaminates the years just before entry into force, Nagengast and Yotov (2025) date the onset **three years before** the agreement legally enters into force.

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Beyond trade agreements

The same logic applies to any staggered gravity policy — WTO accession, currency unions, sanctions — and to gravity equations for migration or FDI.

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